

FED3 Analyzer

Quick Start Guide

A concise guide to Time Bins and Trial Bins workflows for FED3 behavioural data.

1. Getting started

When FED3 Analyzer opens, select either Time Bins or Trial Bins. Time Bins analyses events within fixed time intervals, whereas Trial Bins analyses events according to task-specific behavioural trials.

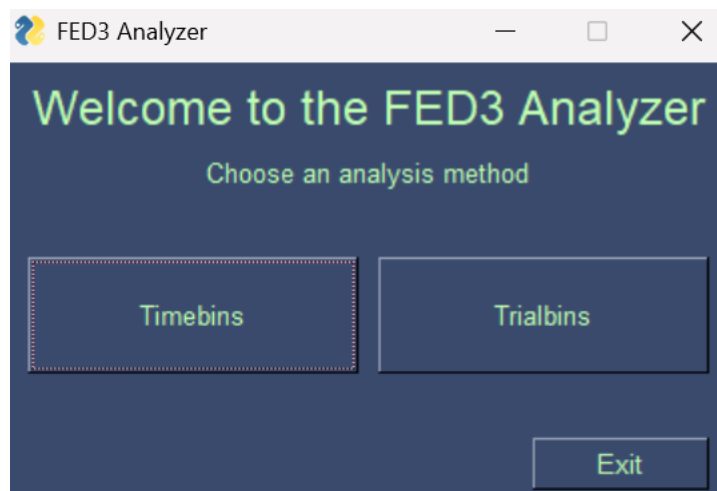


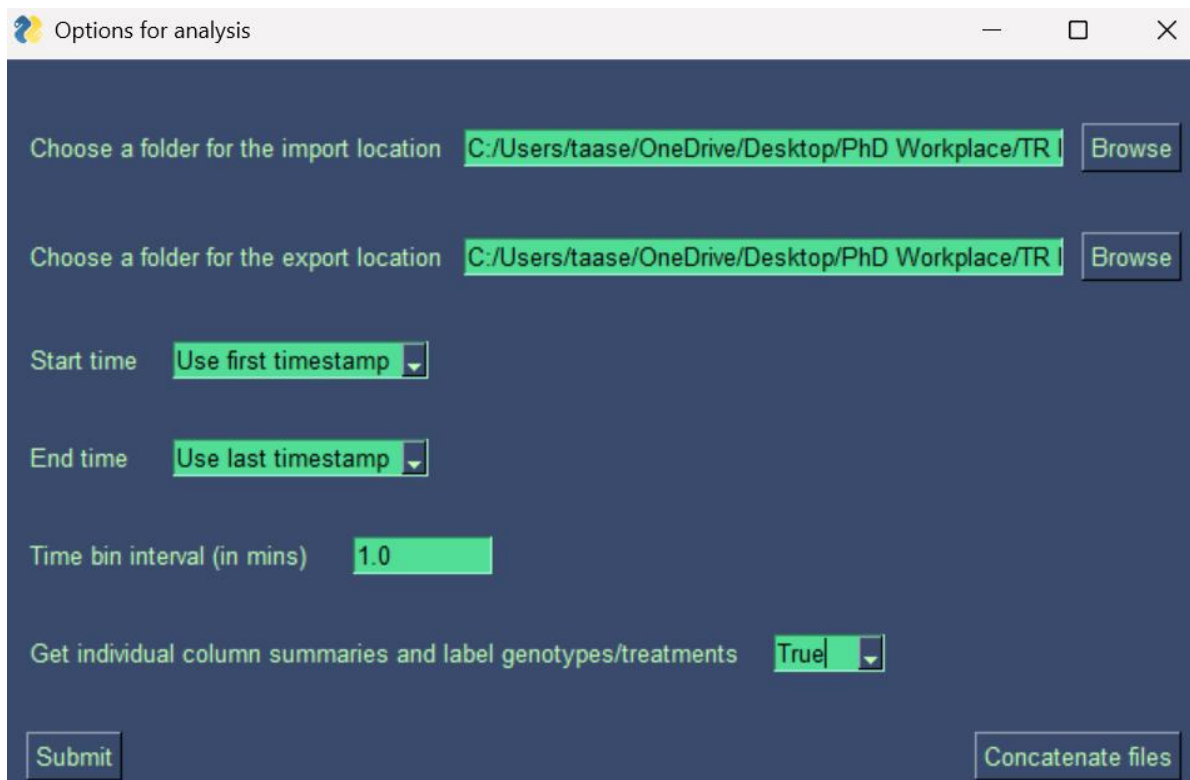
Figure 1. FED3 Analyzer analysis selection window.

2. Time Bins

Time Bins converts FED3 events into user-defined intervals and can produce individual outputs, combined master workbooks, summary metrics, and plots.

2.1 Select files and analysis settings

Select the import folder containing the raw FED3 CSV files and the export folder where analysed files should be saved. Define the analysis start and end times, time-bin duration, and whether measurements should be combined into master workbooks.



Options for analysis

Choose a folder for the import location

Choose a folder for the export location

Start time

End time

Time bin interval (in mins)

Get individual column summaries and label genotypes/treatments

Figure 2. Initial Time Bins file and analysis settings.

2.2 Long-term session options

Long-term session types—including Closed Economy PR1, Bandit, Stop Signal, and LeftRight—produce task-specific master workbooks. Select True under “Create summed master Excel files?” to generate cumulative versions of these workbooks.

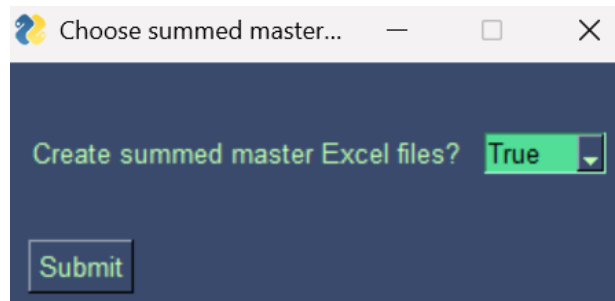


Figure 3. Option to create cumulative master workbooks.

Long-term sessions can also be divided according to the light/dark cycle. Enter the light-cycle start and end times when prompted; the corresponding dark-cycle period is filled automatically.

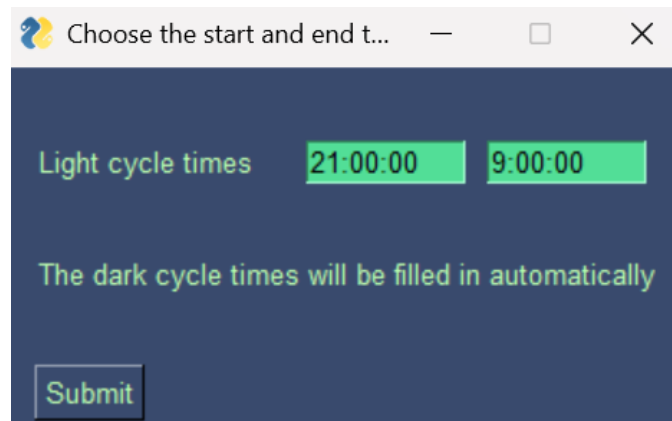
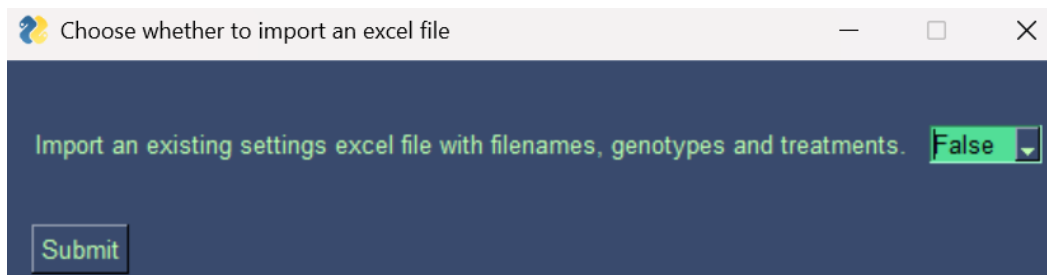


Figure 4. Light-cycle time entry for long-term sessions.

2.3 Animal metadata

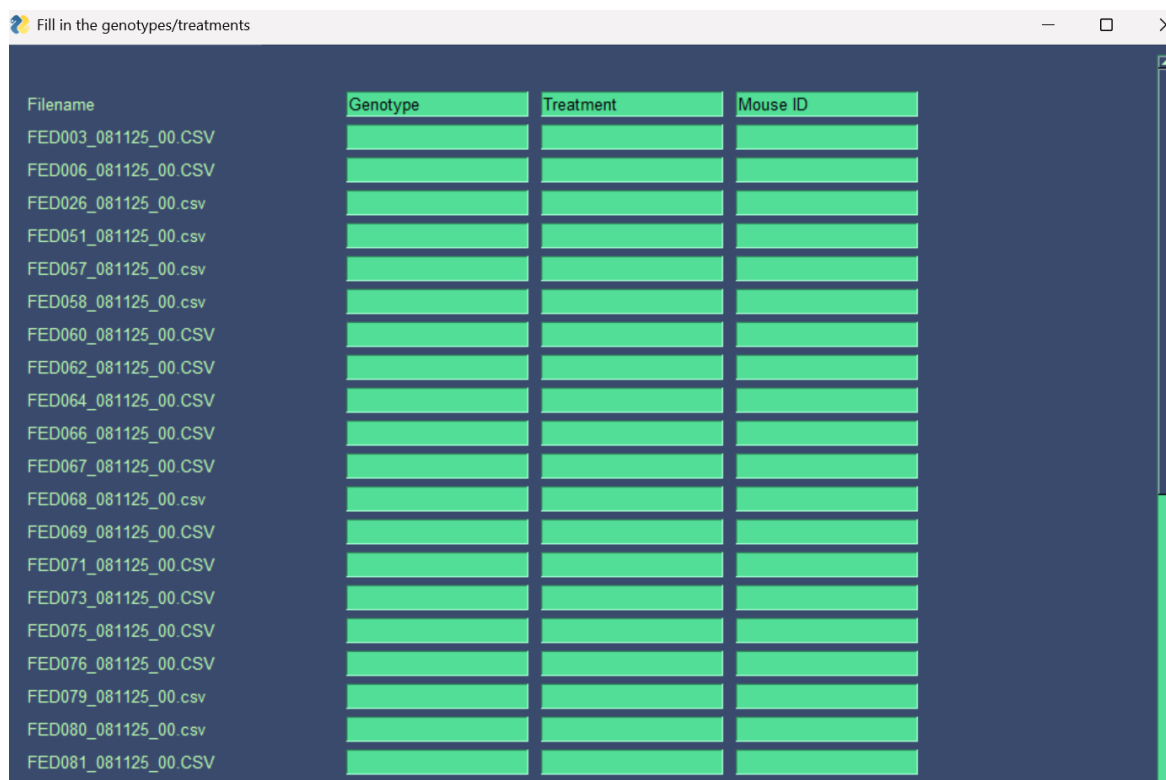
Import an existing settings workbook containing animal metadata or enter the metadata manually. When metadata are entered manually, the column headers can be renamed to suit the experimental design.



Choose whether to import an excel file

Import an existing settings excel file with filenames, genotypes and treatments.

Figure 5. Option to import an existing metadata workbook.

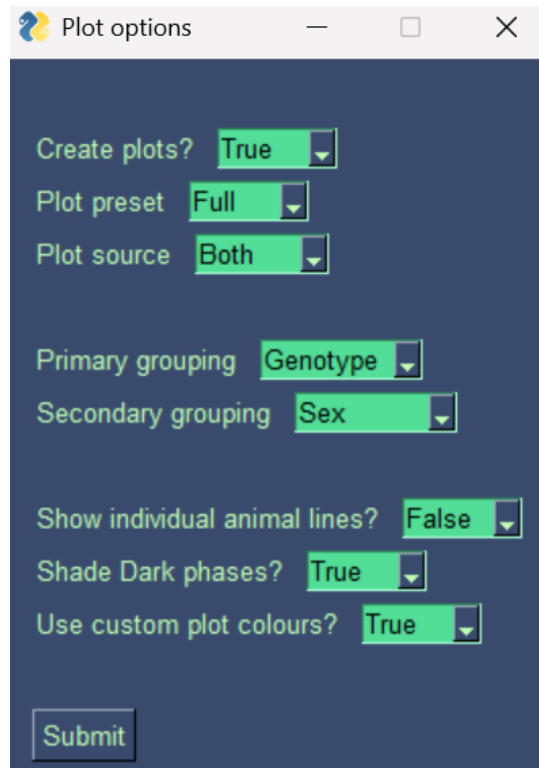


Filename	Genotype	Treatment	Mouse ID
FED003_081125_00.CSV			
FED006_081125_00.CSV			
FED026_081125_00.csv			
FED051_081125_00.csv			
FED057_081125_00.csv			
FED058_081125_00.csv			
FED060_081125_00.CSV			
FED062_081125_00.CSV			
FED064_081125_00.CSV			
FED066_081125_00.CSV			
FED067_081125_00.CSV			
FED068_081125_00.csv			
FED069_081125_00.CSV			
FED071_081125_00.CSV			
FED073_081125_00.CSV			
FED075_081125_00.CSV			
FED076_081125_00.CSV			
FED079_081125_00.csv			
FED080_081125_00.csv			
FED081_081125_00.CSV			

Figure 6. Manual entry of filenames and animal metadata.

2.4 Long-term plotting

Plots can be generated from standard master workbooks, cumulative master workbooks, or both. Primary and secondary grouping variables can be used to create plots for each grouping variable and their interaction.



Plot options

Create plots? **True**

Plot preset **Full**

Plot source **Both**

Primary grouping **Genotype**

Secondary grouping **Sex**

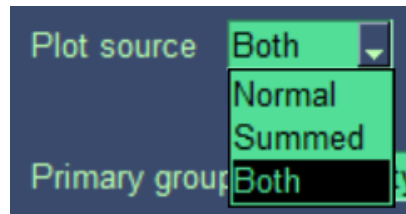
Show individual animal lines? **False**

Shade Dark phases? **True**

Use custom plot colours? **True**

Submit

Figure 7. Long-term plotting options.



Plot source **Both**

Normal

Summed

Primary group **Both**

Figure 8. Plot-source choices: standard, cumulative, or both.

Custom colours can be selected for the generated plots.

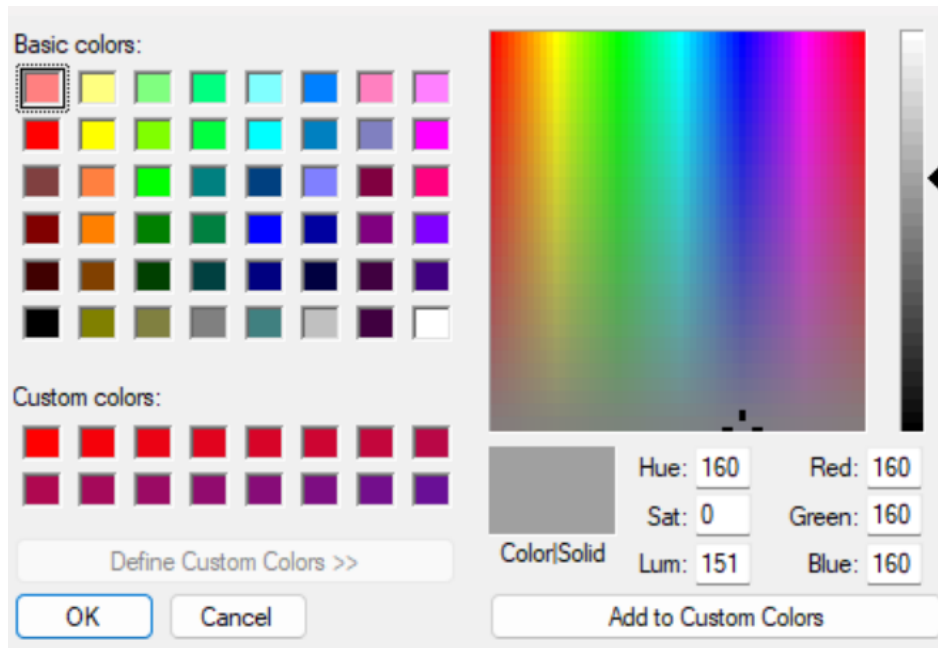


Figure 9. Custom plot-colour selection.

2.5 Short-term session options

Short-term session types—including FR1, FR3, FR5, progressive ratio, and extinction—provide additional options for calculating conventional FED3 metrics and standardising session duration.

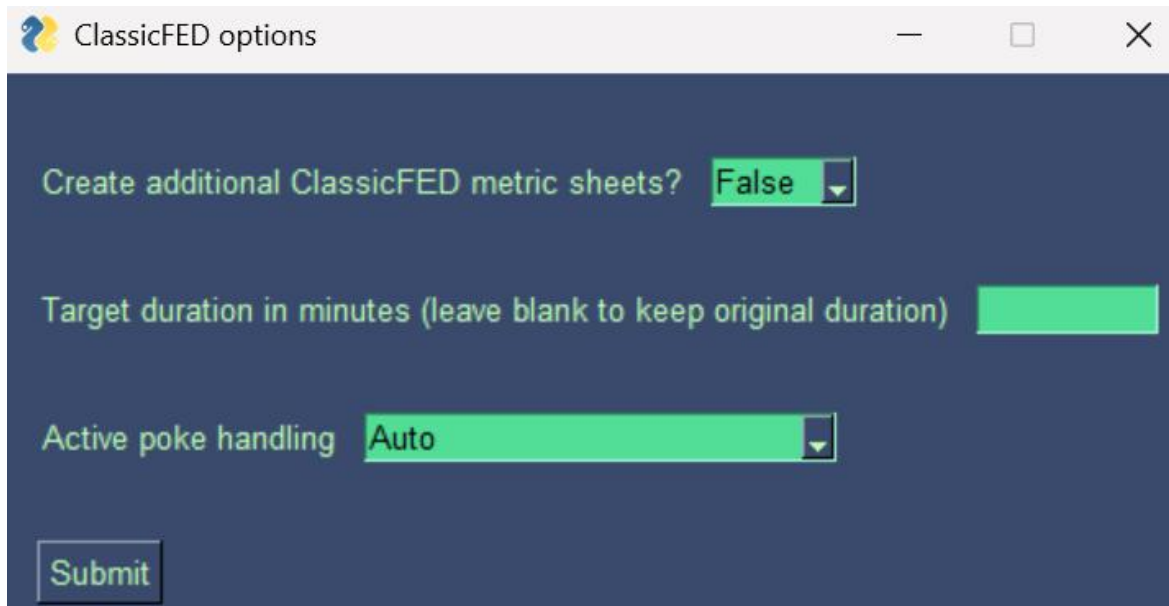
A screenshot of a software window titled "ClassicFED options". The window has a dark blue background with light blue text. It contains three settings: "Create additional ClassicFED metric sheets?" with a dropdown menu set to "False"; "Target duration in minutes (leave blank to keep original duration)" with an empty text input field; and "Active poke handling" with a dropdown menu set to "Auto". A "Submit" button is located at the bottom left of the dialog.

Figure 10. ClassicFED metrics, target duration, and active-poke settings.

Additional ClassicFED metrics

Select True under “Create additional ClassicFED metric sheets?” to calculate measurements such as response accuracy, pellet-retrieval efficiency, and event averages. These measurements are added to the output master workbooks.

Target duration

Sessions can be cropped or extended to a specified duration. For example, entering 45 standardises every output to 45 minutes, even if individual recordings are slightly shorter or longer.

Active poke handling

- Auto uses the Active_Poke column in the raw FED3 file.
- Left fallback uses the left side when the active side cannot be identified automatically.
- Right fallback uses the right side when the active side cannot be identified automatically.
- Skip active-side metrics excludes measurements that require an active response side.

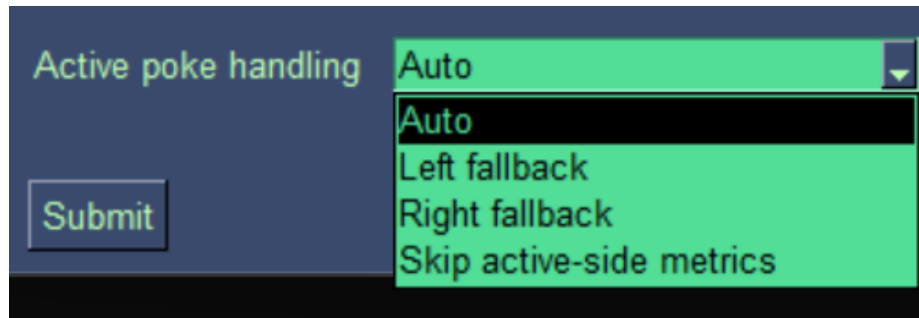


Figure 11. Active-poke handling choices.

Short-term sessions also support grouped plots, individual-animal traces, light/dark shading, and custom colours.

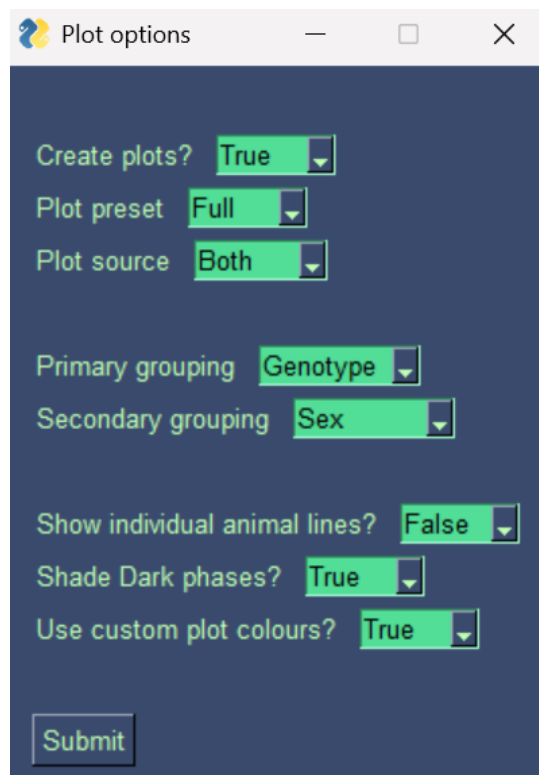


Figure 12. Short-term plotting options.

3. Trial Bins

Trial Bins groups FED3 events according to behavioural trials rather than fixed time intervals. Three task-specific analyses are currently available:

- Closed Economy PR1
- Stop Signal
- Bandit

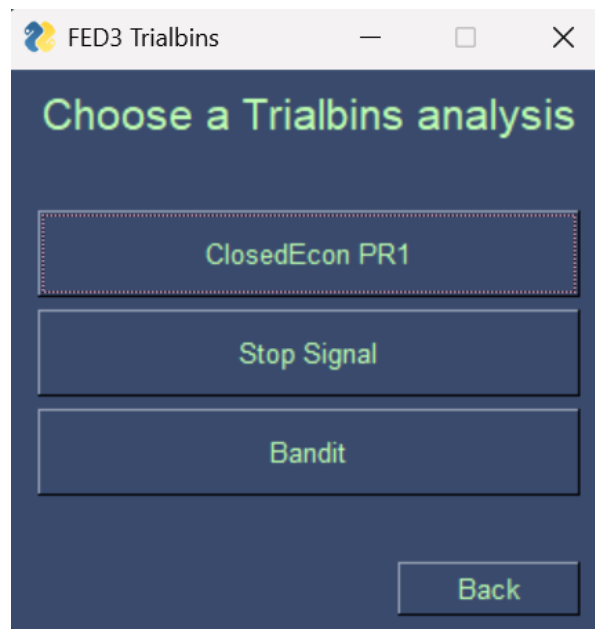


Figure 13. Trial Bins analysis selection window.

3.1 Select input files

Select the required analysis and choose the raw FED3 CSV files when prompted.

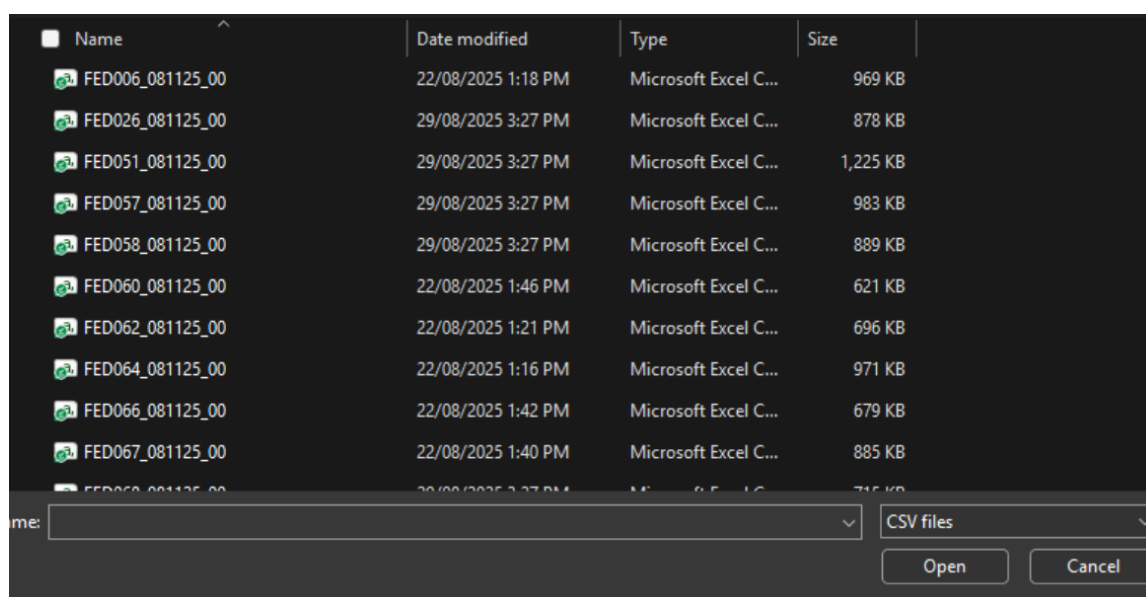


Figure 14. Selection of raw FED3 CSV files.

3.2 Configure the analysis

Depending on the selected analysis, the data can be split by light/dark cycle. Animal metadata can be imported from an existing workbook or entered manually.

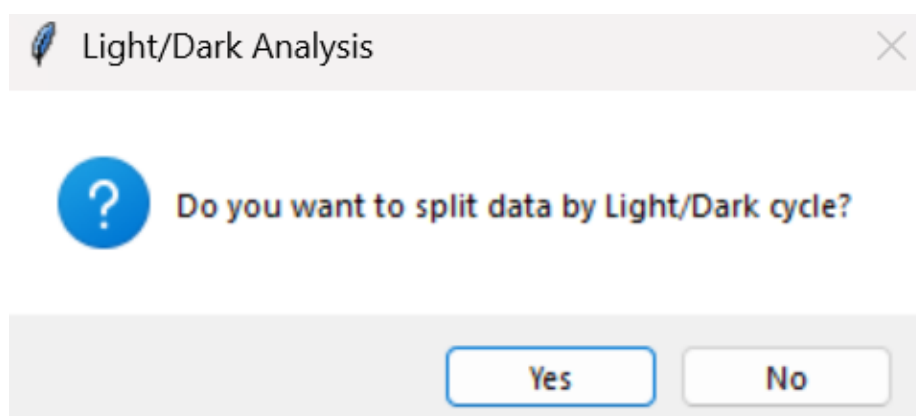


Figure 15. Option to split Trial Bins data by light/dark cycle.

 Enter Metadata

Filename	Mouse ID	Sex	Genotype
FED006_081125_00.CSV			
FED026_081125_00.csv			
FED057_081125_00.csv			
FED058_081125_00.csv			
FED060_081125_00.CSV			
FED062_081125_00.CSV			
FED064_081125_00.CSV			
FED066_081125_00.CSV			
FED067_081125_00.CSV			
FED068_081125_00.csv			
FED069_081125_00.CSV			
FED071_081125_00.CSV			
FED073_081125_00.CSV			
FED075_081125_00.CSV			
FED076_081125_00.CSV			
FED079_081125_00.csv			
FED080_081125_00.csv			
FED081_081125_00.CSV			
FED082_081125_00.CSV			
FED083_081125_00.CSV			
FED084_081125_00.csv			
FED085_081125_00.CSV			
FED087_081125_00.CSV			

Figure 16. Manual entry of Trial Bins metadata.

Custom plot colours can be selected before processing begins.

3.3 Outputs

After processing, the program saves the generated plots and an Excel workbook ending in _EXTRAS.xlsx. This workbook contains the analysed data used for plotting and can also be imported into software such as GraphPad Prism.

4. Further guidance

This quick-start guide introduces the main analysis workflow. Refer to the task-specific documentation supplied with FED3 Analyzer for detailed information about inputs, analysis rules, output worksheets, and interpretation.

- Time Bins: FED3_time_bins/README.md and FED3_time_bins/How_to_use_FED_code.pdf
- Closed Economy PR1 Trial Bins: FED3_trial_bins/ClosedEconPR1_README.txt
- Stop Signal Trial Bins: FED3_trial_bins/StopSig_README.txt
- Bandit Trial Bins: FED3_trial_bins/Bandit_README.txt

FED3 Analyzer is intended to support reproducible processing of FED3 behavioural data. Review all selected settings and validate representative outputs before applying a workflow to a complete dataset.